

Hamza Khan

Senior Backend Engineer · Python · FastAPI · Kafka · AWS · Distributed Systems

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SUMMARY

Backend engineer with 5+ years shipping production Python services at scale. Drove a 30% reduction in database response time, cut peak-traffic queue latency by 40%, and shipped AI-powered pipelines serving 50M+ users. Specializes in microservices, event-driven architectures (Kafka), RESTful services, and AWS. Strong advocate of code review, automated testing, and small-batch continuous delivery as the path to reliability.

EXPERIENCE

Senior Software Engineer · [Techolution](#)

Jun. 2025 – Present | Remote

- Architected Python microservices for a high-scale audio streaming & podcast platform (iHeartMedia) serving 50M+ monthly active users across 850+ radio stations, delivered via small-batch, CI-driven releases.
- Shipped an AI-powered document ingestion pipeline (Python, Celery, Redis, multi-agent LLM) that cut manual processing time by 70% and onboards 10K+ documents/month.
- Diagnosed and fixed async context-propagation bugs causing metadata inconsistency in real-time podcast pipelines, driving error rate from 15% to under 1%.
- Designed Redis-backed failure-tracking for LLM API workflows, reducing undetected task failures by 90% and on-call alerts by 30%.
- Owned event-driven Kafka pipelines processing 100K+ real-time audio events per hour with sub-second end-to-end latency.
- Led SQL and backend workflow optimizations contributing to a 30% reduction in average database response time across core services, validated through unit and integration tests in CI.

Software Engineer 2 · [Xempla, Enterprise Asset Management Platform](#)

Nov. 2021 – Jun. 2025 | Remote (Australia)

- Scaled backend services and event-driven data pipelines for enterprise analytics workloads across 200+ client sites globally, working with both relational (PostgreSQL) and non-relational (MongoDB) stores.
- Re-architected the data pipeline to reduce infrastructure load by 35%, lowering cloud compute costs by an estimated \$80K/year.
- Delivered analytics systems and RESTful API integrations that accelerated a new product-line launch by 3 months and onboarded 15+ enterprise customers in year one.
- Resolved 30+ complex production incidents while maintaining a 99.7% uptime SLA across all managed pipelines.
- Introduced code-review standards and expanded automated test coverage, cutting post-deployment bug reports by 25%; mentored 3 junior engineers on the same practices.

Software Engineer · [WatchMyDC Analytics](#)

Mar. 2021 – Nov. 2021 | Remote (Finland)

- Built observability tooling (metrics, alerts, dashboards) that cut mean time to detection from 45 minutes to under 10 minutes.
- Developed scalable Django REST APIs powering CRM systems for 5K+ daily active users at <200 ms p95 response time.

Web Developer · [Croudit](#)

Oct. 2020 – Mar. 2021 | Remote

- Designed server-side architecture in Python (Django & Flask), delivering 4 backend modules used across 2 product lines.
- Built infrastructure tooling that automated 3 manual deployment steps, saving the team 5 hours/week.

PROJECTS

Kynstro | *Python, FastAPI, OpenAI, Redis, PostgreSQL, AWS*

kynstro.com

- Designed and shipped an end-to-end AI interview platform as the sole backend engineer, from system design to deployment.
- Built async LLM-powered interview pipelines, real-time scoring logic, and a multi-tenant API layer on FastAPI.
- Owned the full stack: schema design, queueing, LLM orchestration, observability, and AWS deployment.

TECHNICAL SKILLS

Languages: Python (primary), JavaScript

Backend: FastAPI, Django, Flask, REST APIs, Microservices, Async Programming

Messaging & Data: Apache Kafka, Redis, Celery, PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS (EC2, S3, SQS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, Trunk-Based Development

Testing & Quality: Unit Testing, Integration Testing, Code Review, Automated CI Pipelines

Architecture: Distributed Systems, Event-Driven Design, RESTful Services, System Design, Scalable Services

AI / LLM: OpenAI APIs, AI Agents, LLM Integrations, Multi-Agent Pipelines

EDUCATION

Integral University — Lucknow, India

Bachelor in Computer Application · CGPA: 8.9/10

2020 – 2023